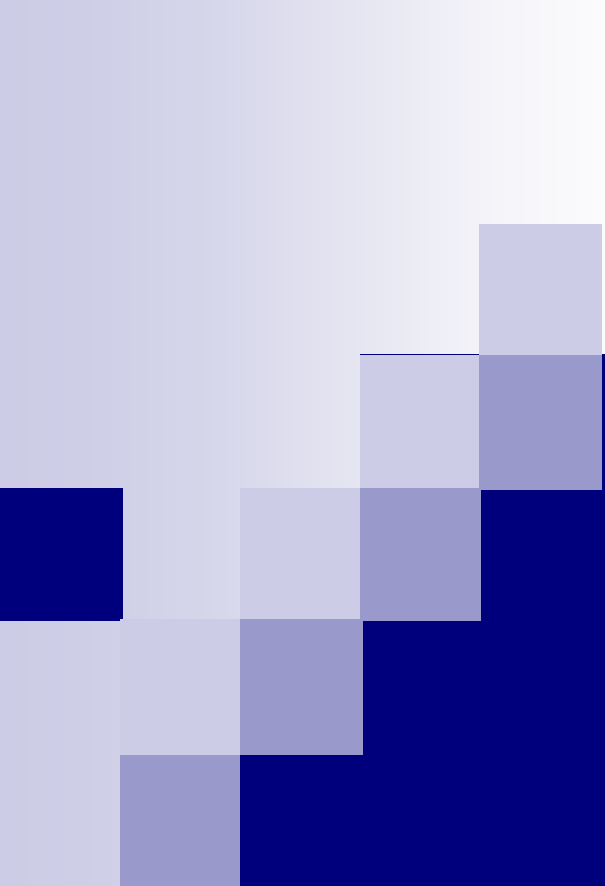




Chapter 8

Random-Variate Generation

Banks, Carson, Nelson & Nicol
Discrete-Event System Simulation

Purpose & Overview

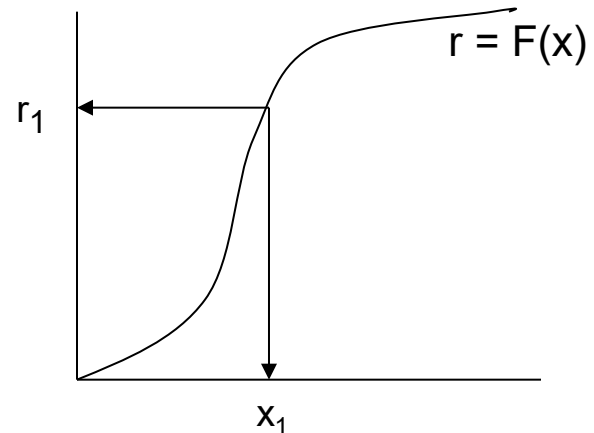
- Develop understanding of generating samples from a specified distribution as input to a simulation model.
- Illustrate some widely-used techniques for generating random variates.
 - Inverse-transform technique
 - Acceptance-rejection technique
 - Special properties

Inverse-transform Technique

- The concept:

- For cdf function: $r = F(x)$
- Generate r from uniform $(0,1)$
- Find x :

$$x = F^{-1}(r)$$



Steps in inverse-transform technique

Step 1. Compute the cdf of the desired random variable X : $F(x) = 1 - e^{-\lambda x}$

$$x \geq 0$$

Step 2. Set $F(X) = R$ on the range of X

Step 3. Solve the equation $F(x) = R$ for X in terms of R .

$$1 - e^{-\lambda X} = R$$

$$e^{-\lambda X} = 1 - R$$

$$-\lambda X = \ln(1 - R)$$

$$X = -\frac{1}{\lambda} \ln(1 - R)$$

Step 4. Generate (as needed) uniform random numbers $R_1, R_2, R_3, \dots$ and compute the desired random variates

See the next slide

Exponential Distribution

[Inverse-transform]

■ Exponential Distribution:

□ Exponential cdf:

$$\begin{aligned} r &= F(x) \\ &= 1 - e^{-\lambda x} \quad \text{for } x \geq 0 \end{aligned}$$

□ To generate $X_1, X_2, X_3 \dots$

$$\begin{aligned} X_i &= F^{-1}(R_i) \\ &= -(1/\lambda) \ln(1-R_i) \quad [\text{Eq'n 8.3}] \end{aligned}$$

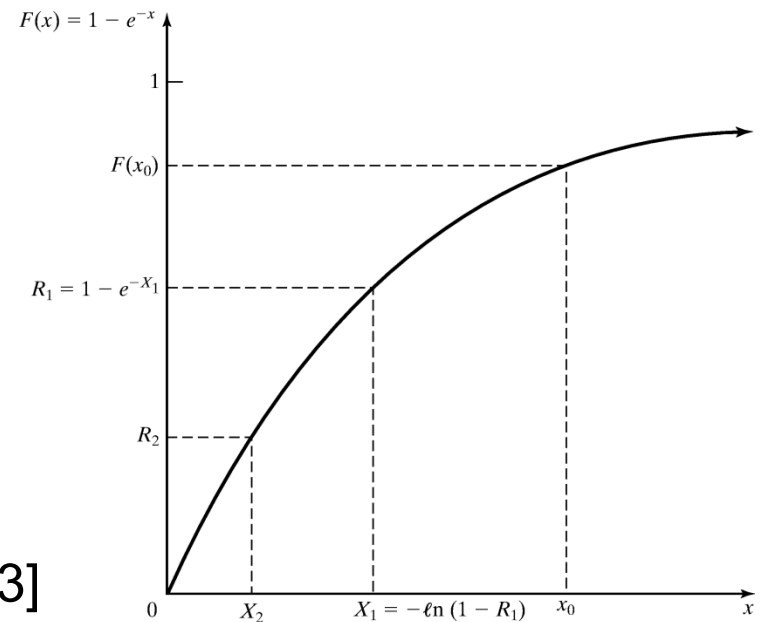
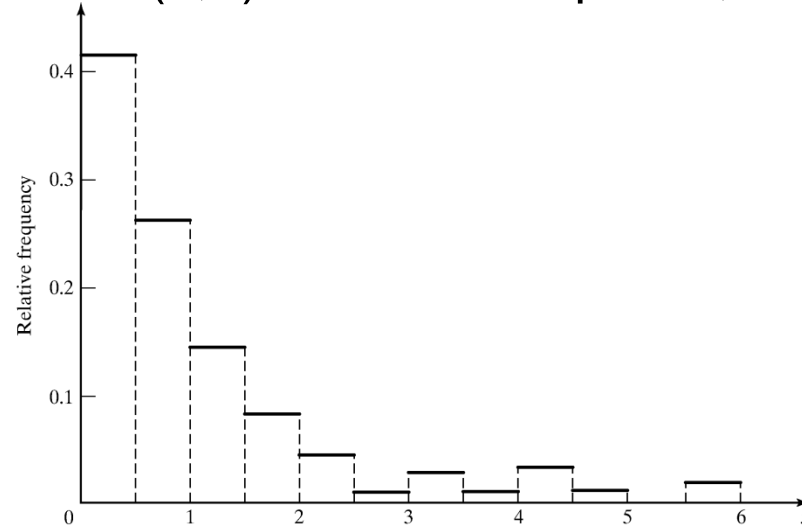


Figure: Inverse-transform technique for $\exp(\lambda = 1)$

Exponential Distribution

[Inverse-transform]

- Example: Generate 200 variates X_i with distribution $\exp(\lambda = 1)$
 - Generate 200 R s with $U(0,1)$ and utilize eq'n 8.3, the histogram of X s become:



- Check: Does the random variable X_1 have the desired distribution?

$$P(X_1 \leq x_0) = P(R_1 \leq F(x_0)) = F(x_0)$$

Does the random variable X_1 have the desired distribution?

- Pick a value of x_0 and compute the cumulative probability.

$$P(X_1 \leq x_0) = P(R_1 \leq F(x_0)) = F(x_0) \quad (8.4)$$

- First equality: See figure 8.2 on slide 5.
- It can be seen that $X_1 \leq x_0$ when and only when $R_1 \leq F(x_0)$.
- Since $0 \leq F(x_0) \leq 1$, the second equality in the equation follows immediately from the fact that R_1 is uniformly distributed on $[0,1]$.
- The equation shows that the cdf of X_1 is F ;
 - hence X_1 has the desired distribution

Other Distributions

[Inverse-transform]

- Examples of other distributions for which inverse cdf works are:

- Uniform distribution

$$X = a + (b - a)R$$

- Weibull distribution – time to failure – see steps on p278

$$X = \alpha[-\ln(1 - R)]^{1/\beta}$$

- Triangular distribution

$$X = \begin{cases} \sqrt{2R}, & 0 \leq R \leq 1/2 \\ 2 - \sqrt{2(1-R)}, & 1/2 < R \leq 1 \end{cases}$$

Section 8.1.5 Empirical Continuous Distributions



- This is a worthwhile read (as is the whole chapter of course)
- Works on the question:
 - What do you do if you can't figure out what the distribution of the data is?
- The example starting on slide 13 is a good model to work from.

Empirical Continuous Dist'n [Inverse-transform]

- When theoretical distribution is not applicable
- To collect empirical data:
 - Resample the observed data (i.e. use the data for the distribution)
 - Interpolate between observed data points to fill in the gaps
- For a small sample set (size n):
 - Arrange the data from smallest to largest

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$$

- Assign the probability $1/n$ to each interval $x_{(i-1)} \leq x \leq x_{(i)}$

$$X = \hat{F}^{-1}(R) = x_{(i-1)} + a_i \left(R - \frac{(i-1)}{n} \right)$$

where
$$a_i = \frac{x_{(i)} - x_{(i-1)}}{1/n - (i-1)/n} = \frac{x_{(i)} - x_{(i-1)}}{1/n}$$

Empirical Continuous Dist'n [Inverse-transform]

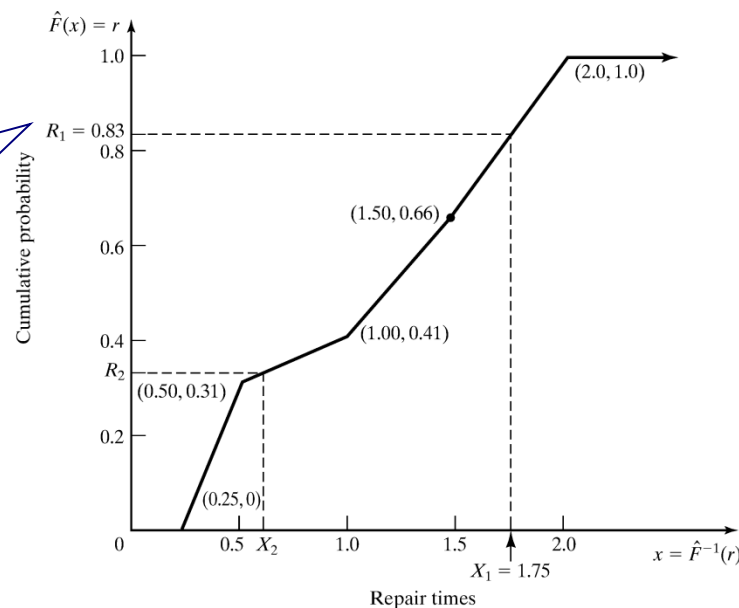
- Example: Suppose the data collected for 100 broken-widget repair times are:

i	Interval (Hours)	Frequency	Relative Frequency	Cumulative Frequency, c_i	Slope, a_i
1	$0.25 \leq x \leq 0.5$	31	0.31	0.31	0.81
2	$0.5 \leq x \leq 1.0$	10	0.10	0.41	5.0
3	$1.0 \leq x \leq 1.5$	25	0.25	0.66	2.0
4	$1.5 \leq x \leq 2.0$	34	0.34	1.00	1.47

Consider $R_1 = 0.83$:

$$c_3 = 0.66 < R_1 < c_4 = 1.00$$

$$\begin{aligned} X_1 &= x_{(4-1)} + a_4(R_1 - c_{(4-1)}) \\ &= 1.5 + 1.47(0.83 - 0.66) \\ &= 1.75 \end{aligned}$$



8.1.6

- There are continuous distributions without a nice closed-form expression for their cdf or its inverse.
 - Normal distribution
 - Gamma
 - Beta
- Must approximate in these cases

Discrete Distribution

[Inverse-transform]

- All discrete distributions can be generated via inverse-transform technique
- Method: numerically, table-lookup procedure, algebraically, or a formula
- Examples of application:
 - Empirical
 - Discrete uniform
 - Gamma

Example 8.4

An Empirical Discrete Distribution

[Inverse-transform]

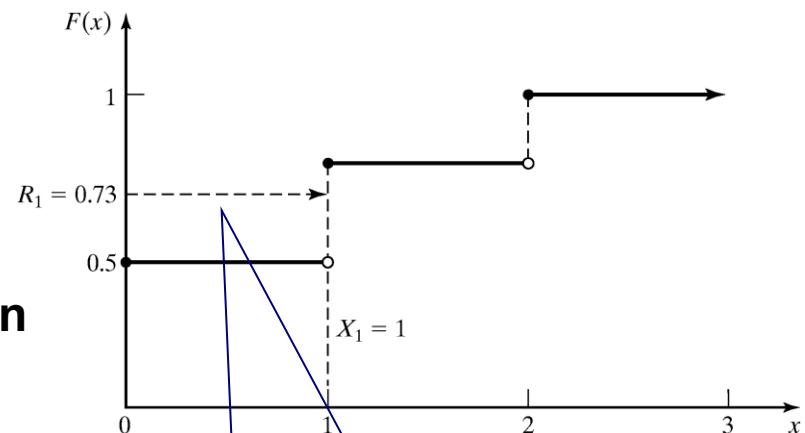
- Example: Suppose the number of shipments, x , on the loading dock of IHW company is either 0, 1, or 2

- Data - Probability distribution:

x	$p(x)$	$F(x)$
0	0.50	0.50
1	0.30	0.80
2	0.20	1.00

- Method - **Given R , the generation scheme becomes:**

$$x = \begin{cases} 0, & R \leq 0.5 \\ 1, & 0.5 < R \leq 0.8 \\ 2, & 0.8 < R \leq 1.0 \end{cases}$$



Consider $R_1 = 0.73$:

$$F(x_{i-1}) < R \leq F(x_i)$$

$$F(x_0) < 0.73 \leq F(x_1)$$

Hence, $x_1 = 1$

Discrete distributions continued



- Example 8.5 concerns a Discrete Uniform Distribution
- Example 8.6 concerns the Geometric Distribution

8.2: Acceptance-Rejection technique

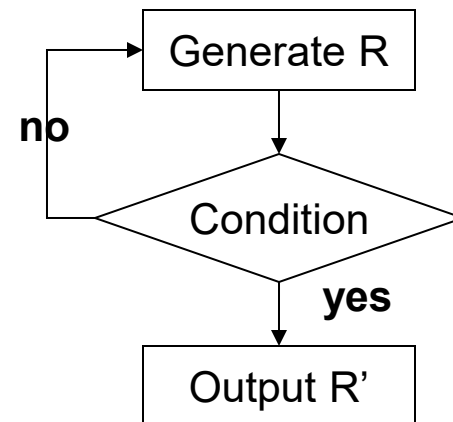
- Useful particularly when inverse cdf does not exist in closed form, a.k.a. thinning
- Illustration: To generate random variates, $X \sim U(1/4, 1)$

Procedures:

Step 1. Generate $R \sim U[0,1]$

Step 2a. If $R \geq 1/4$, accept $X=R$.

Step 2b. If $R < 1/4$, reject R , return to Step 1

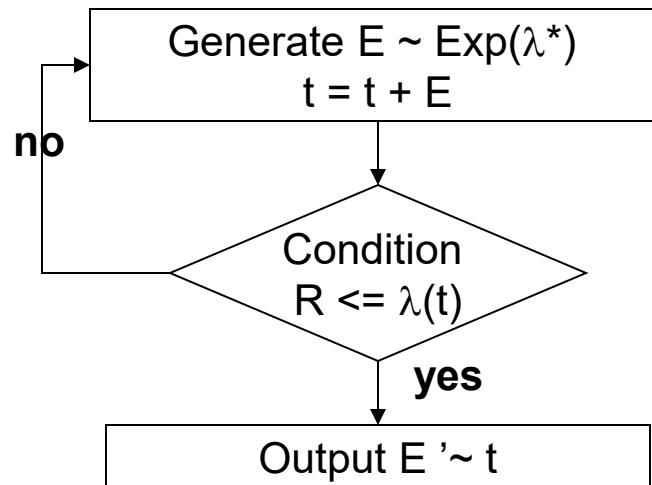


- R does not have the desired distribution, but R conditioned (R') on the event $\{R \geq 1/4\}$ does. (8.21, P. 289)
- Efficiency: Depends heavily on the ability to minimize the number of rejections.

NSPP

[Acceptance-Rejection]

- Non-stationary Poisson Process (NSPP): a Poisson arrival process with an arrival rate that varies with time
- Idea behind thinning:
 - Generate a stationary Poisson arrival process at the fastest rate, $\lambda^* = \max \lambda(t)$
 - But “accept” only a portion of arrivals, thinning out just enough to get the desired time-varying rate



8.2 Acceptance – Rejection continued

■ 8.2.1 Poisson Distribution

□ Step 1 set $n = 0$, $P = 1$

□ Step 2 generate a random number R_{n+1}

And replace P by $P * R_{n+1}$

□ Step 3 if $P < e^{-\lambda}$, then accept, otherwise, reject the current n , increase n by 1 and return to step 2

■ Example: Generate a random variate for a NSPP

Data: Arrival Rates

t (min)	Mean Time Between Arrivals (min)	Arrival Rate $\lambda(t)$ (#/min)
0	15	1/15
60	12	1/12
120	7	1/7
180	5	1/5
240	8	1/8
300	10	1/10
360	15	1/15
420	20	1/20
480	20	1/20

Procedures:

Step 1. $\lambda^* = \max \lambda(t) = 1/5$, $t = 0$ and $i = 1$.

Step 2. For random number $R = 0.2130$,

$$E = -5\ln(0.213) = 13.13$$

$$t = 13.13$$

Step 3. Generate $R = 0.8830$

$$\lambda(13.13)/\lambda^* = (1/15)/(1/5) = 1/3$$

Since $R > 1/3$, do not generate the arrival

Step 2. For random number $R = 0.5530$,

$$E = -5\ln(0.553) = 2.96$$

$$t = 13.13 + 2.96 = 16.09$$

Step 3. Generate $R = 0.0240$

$$\lambda(16.09)/\lambda^* = (1/15)/(1/5) = 1/3$$

Since $R < 1/3$, $T_1 = t = 16.09$,

and $i = i + 1 = 2$

8.3: Special Properties

- Based on features of particular family of probability distributions
- For example:
 - Direct Transformation for normal and lognormal distributions
 - Convolution
 - Beta distribution (from gamma distribution)

Direct Transformation

[Special Properties]

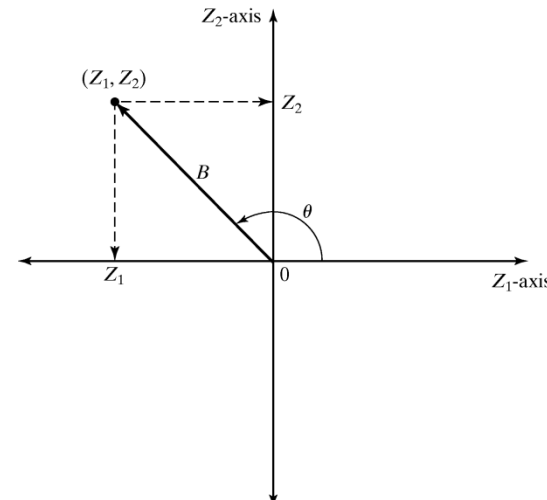
■ Approach for normal(0, 1):

- Consider two standard normal random variables, Z_1 and Z_2 , plotted as a point in the plane:

In polar coordinates:

$$Z_1 = B \cos \phi$$

$$Z_2 = B \sin \phi$$



- $B^2 = Z_1^2 + Z_2^2 \sim$ chi-square distribution with 2 degrees of freedom $= \text{Exp}(\lambda = 2)$. Hence, $B = (-2 \ln R)^{1/2}$
- The radius B and angle ϕ are mutually independent.

$$\begin{aligned} Z_1 &= (-2 \ln R)^{1/2} \cos(2\pi R_2) \\ Z_2 &= (-2 \ln R)^{1/2} \sin(2\pi R_2) \end{aligned}$$

Direct Transformation [Special Properties]

- Approach for normal(μ, σ^2):
 - That is, with mean μ and variance σ^2
 - Generate $Z_i \sim N(0, 1)$ as above

$$X_i = \mu + \sigma Z_i$$

- Approach for lognormal(μ, σ^2):
 - Generate $X \sim N((\mu, \sigma^2))$

$$Y_i = e^{X_i}$$

Summary



- Principles of random-variate generate via
 - Inverse-transform technique
 - Acceptance-rejection technique
 - Special properties
- Important for generating continuous and discrete distributions